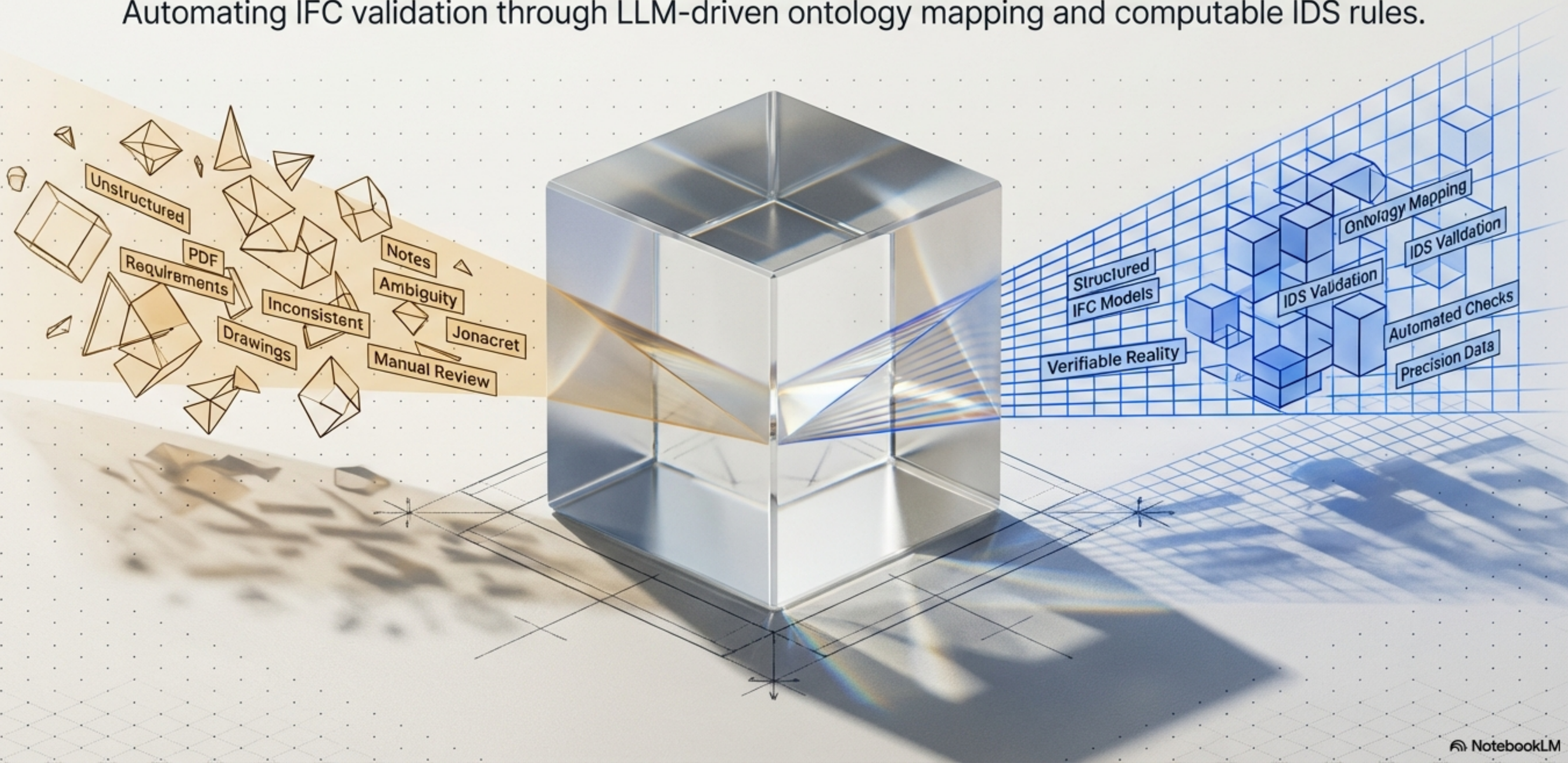


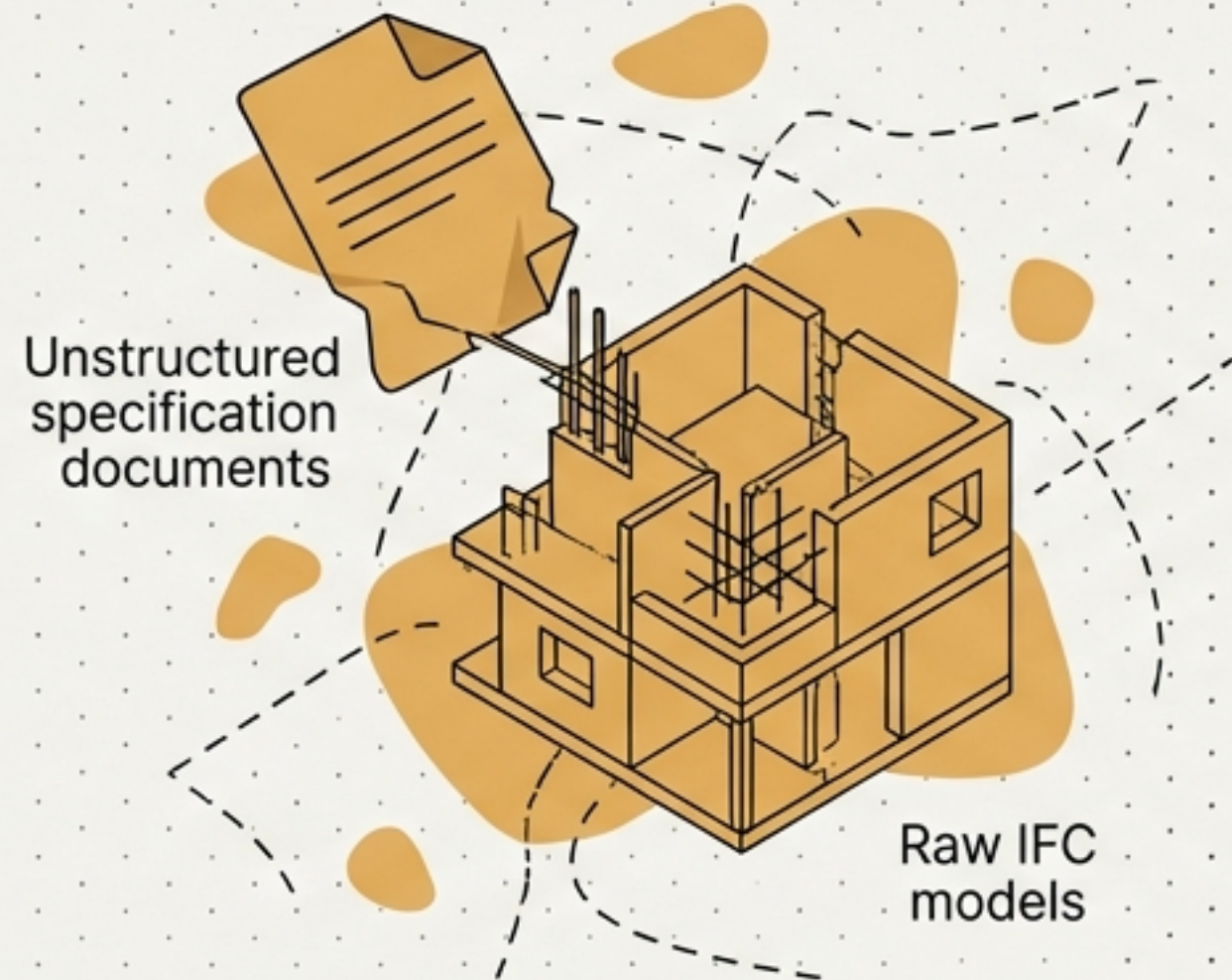
FFELLAA: From Unstructured Requirements to Verifiable Reality

Automating IFC validation through LLM-driven ontology mapping and computable IDS rules.



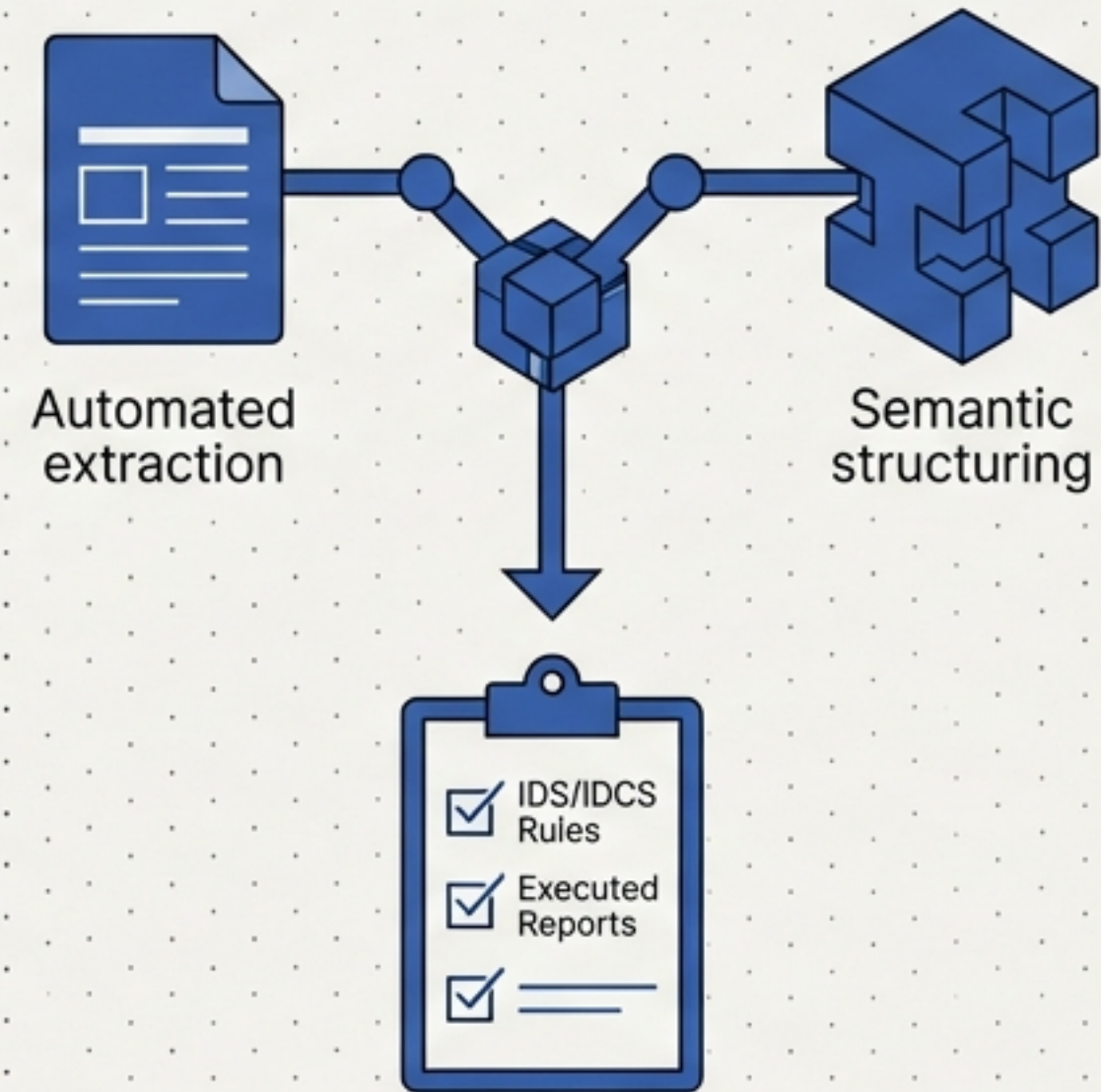
The Synthesis: Bridging Intent and Execution

Disconnected Inputs



Unstructured specification documents and raw IFC models exist in silos, requiring manual translation and verification.

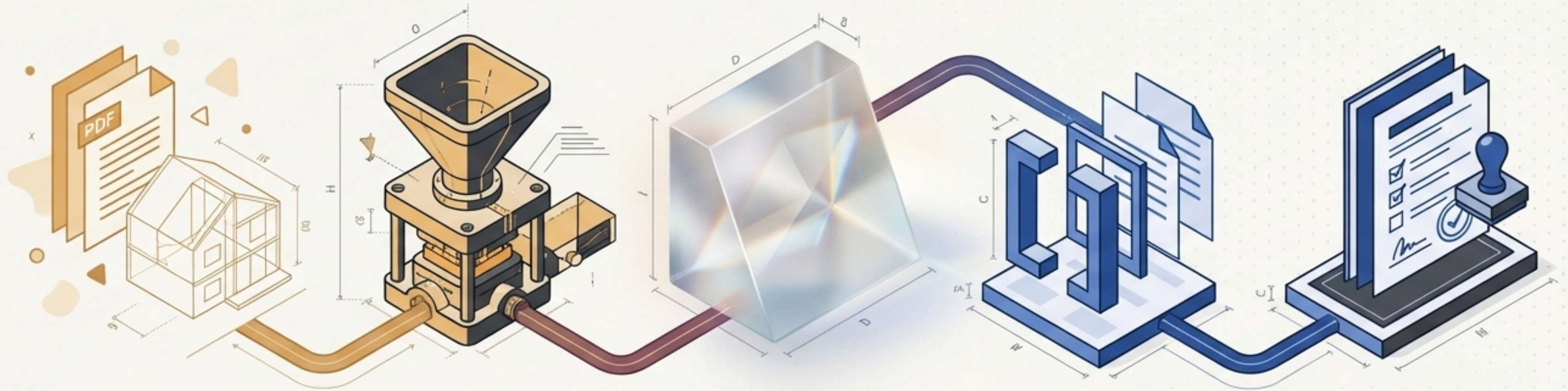
Verifiable Logic



Automated extraction and semantic structuring produce verified rules (IDS/IDCS) and executed reports.

The Macro Pipeline

The FFEELLAA workflow creates a continuous, automated thread from human-readable text to machine-executable structural reports.



01. Inputs

Specification documents (PDF) wireframe and model of raw IFC models.

02. Structuring Tool

Normalizes, extracts, and structures data.

03. LLM & Ontology Engine

Transforms requirements into formal artifacts.

04. Intermediate Outputs

Generates IDS/IDCS for validation rules.

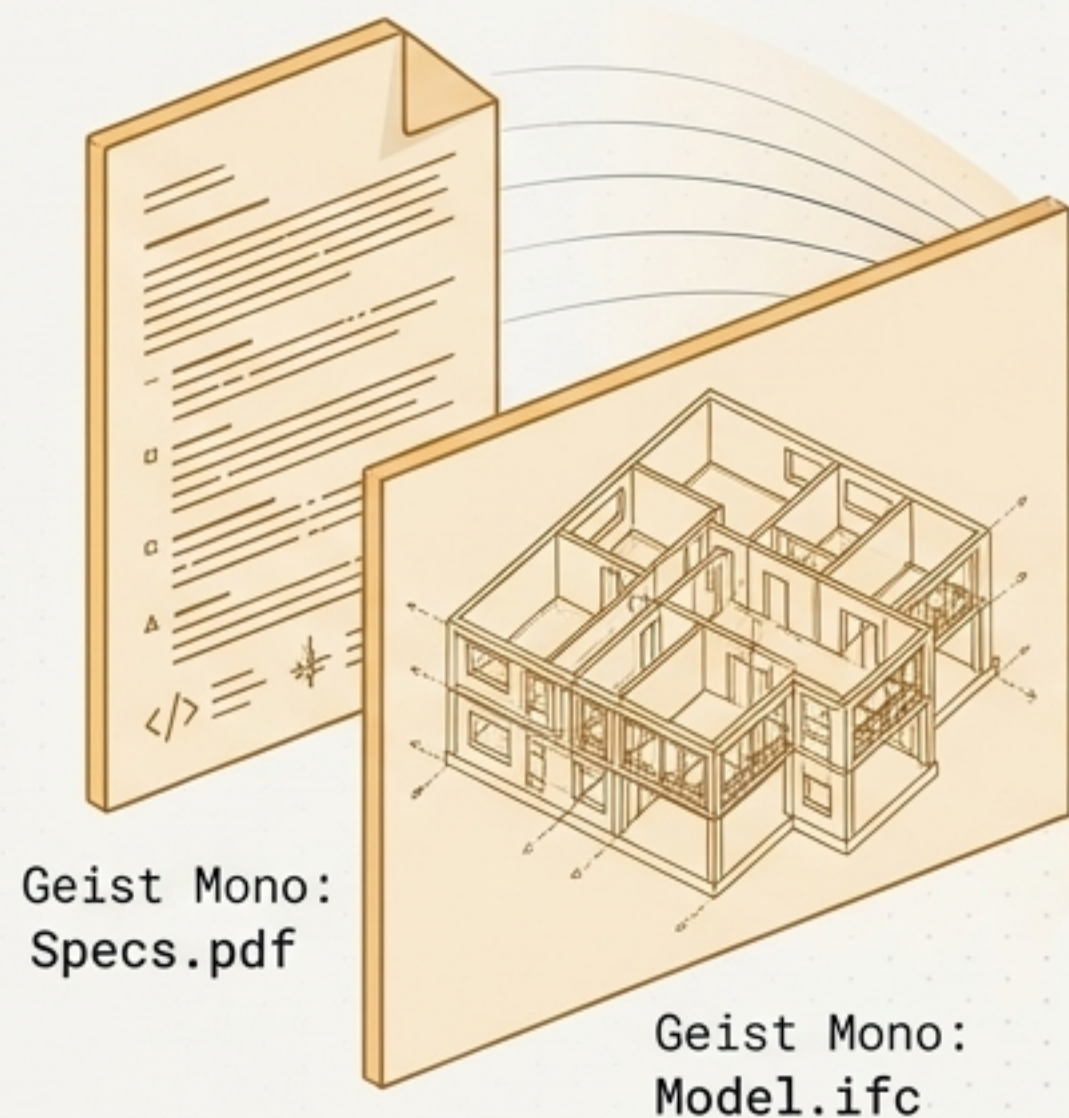
05. Execution Layer

Capabilities consume IDS/IDCS to produce reports.

Ingesting and Normalizing Raw Project Data

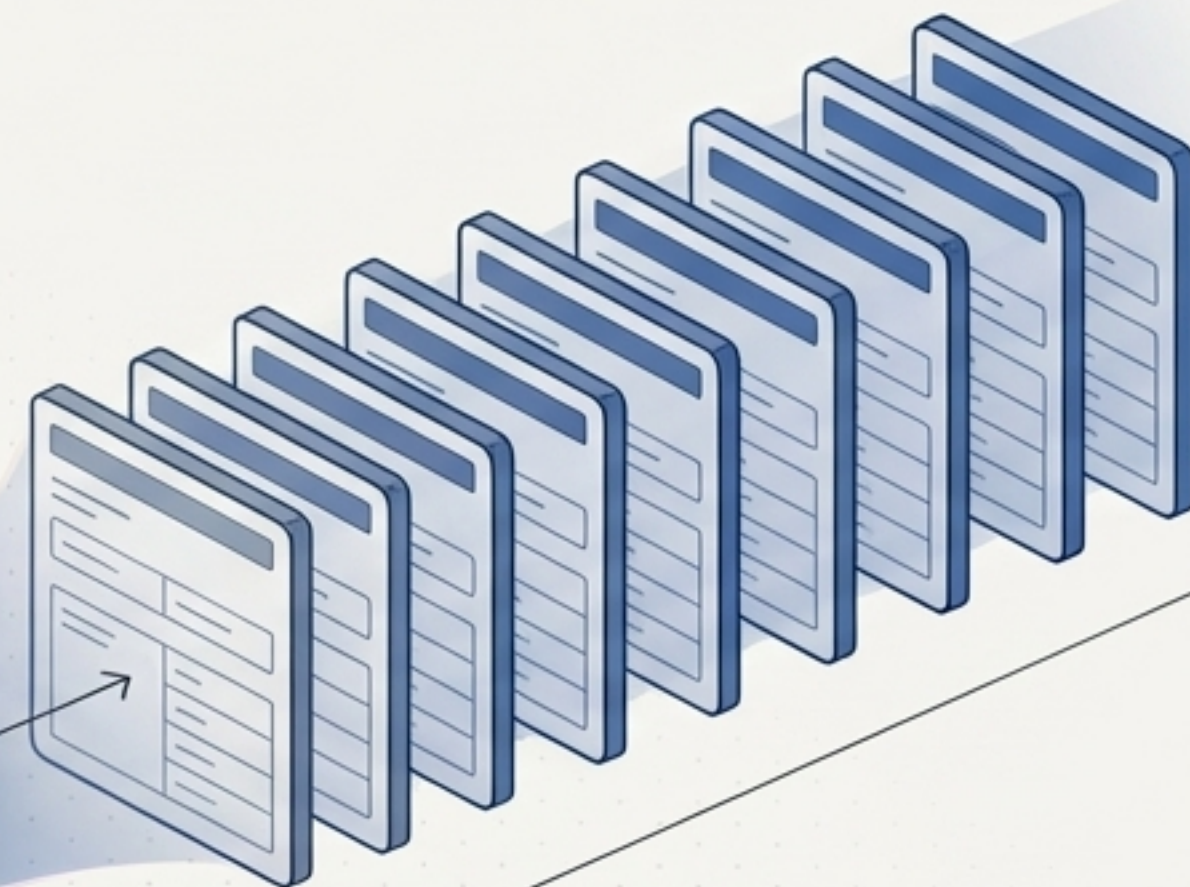
The Input Vector

Ingesting raw architectural specification documents alongside standard industry IFC models.



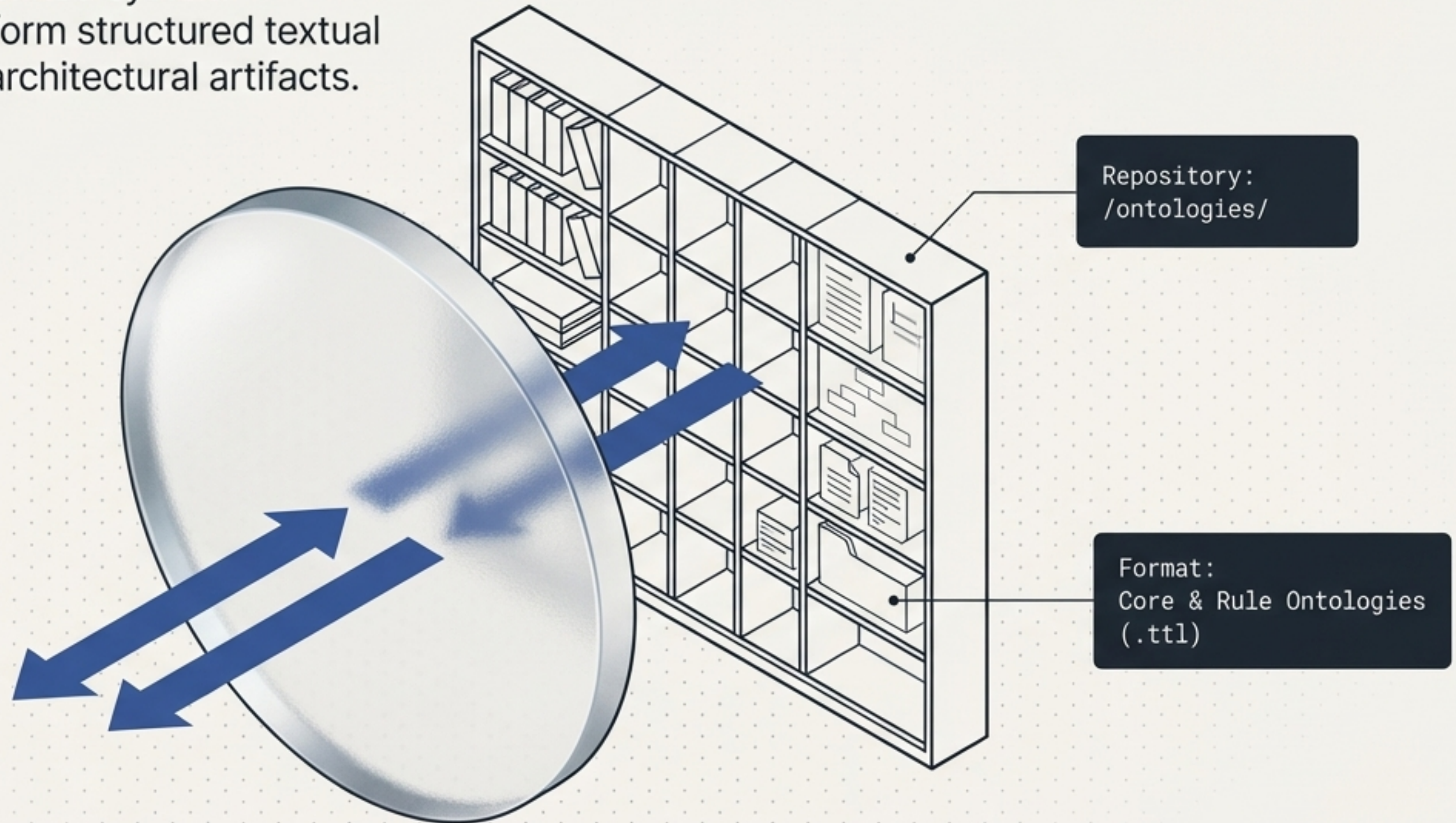
The Structuring Process

The dedicated tool layer normalizes inputs, extracts key parameters, and structures unstructured text into a format optimized for LLM processing.



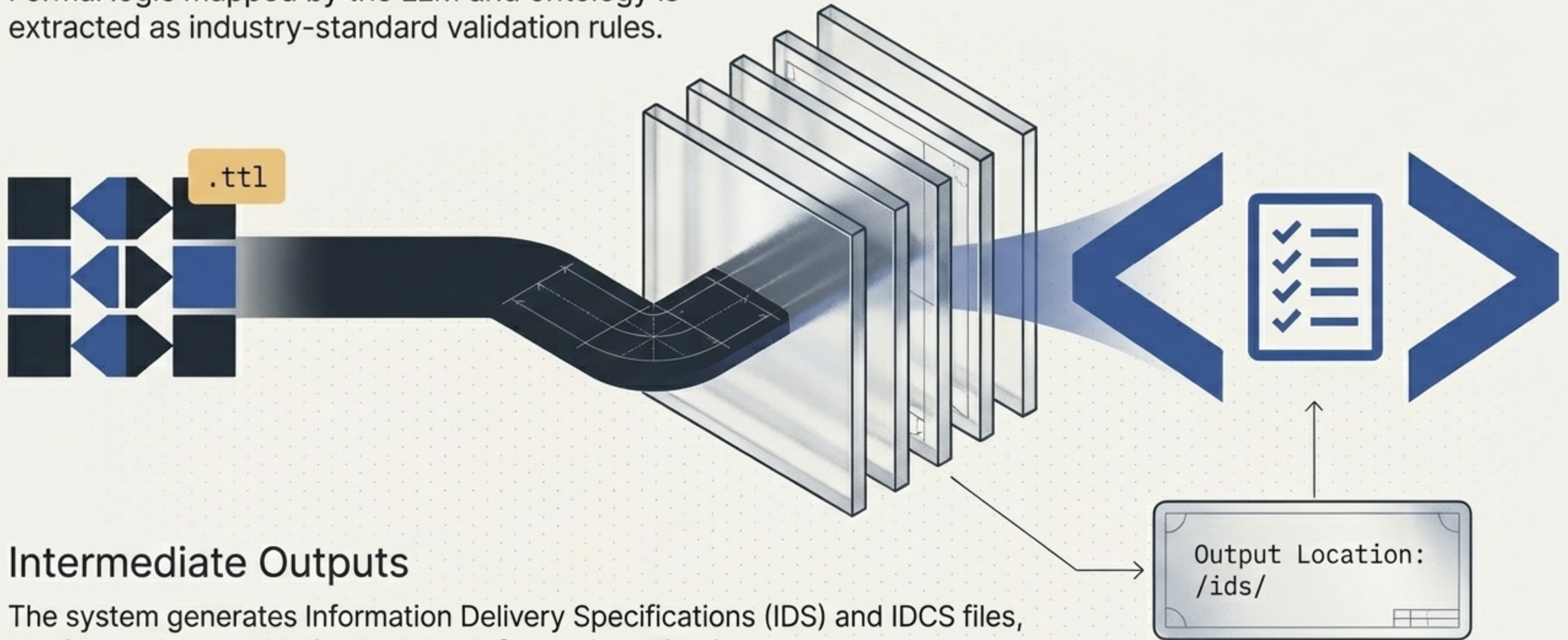
The Semantic Brain: Mapping Intent to Ontology

The LLM uses explicitly defined system ontologies to safely transform structured textual requirements into formal architectural artifacts.



Generating Computable Artifacts

Formal logic mapped by the LLM and ontology is extracted as industry-standard validation rules.

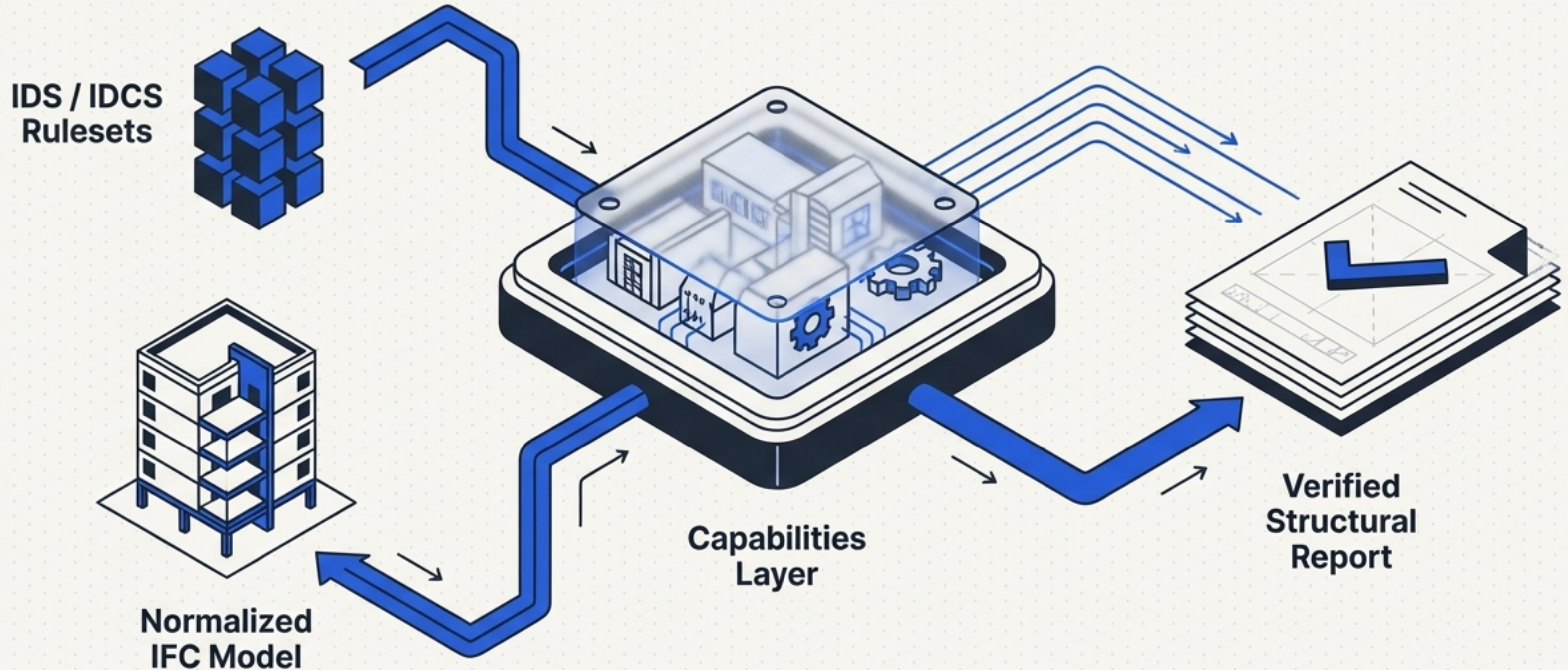


Intermediate Outputs

The system generates Information Delivery Specifications (IDS) and IDCS files, serving as the standardized rulebook for model validation.

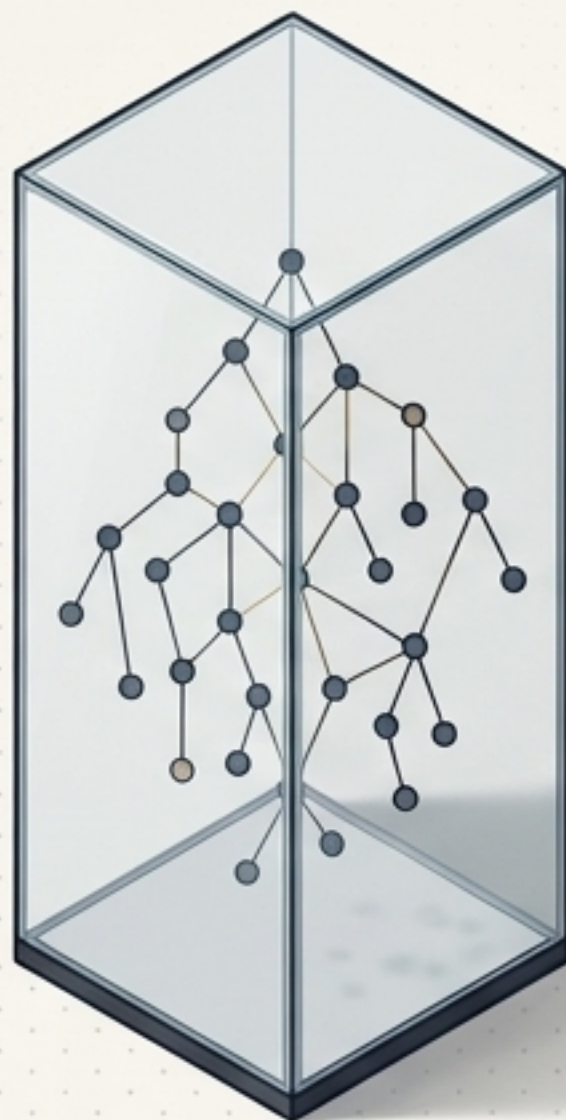
Execution and Reporting

The final pass: computable rules are executed directly against normalized IFC models.



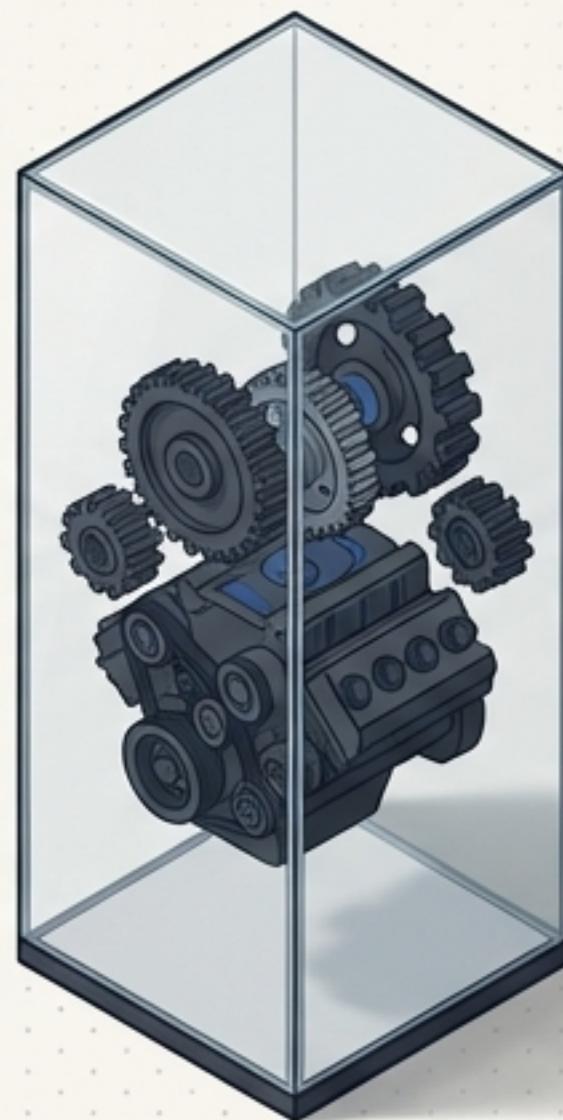
The Component Triad

Three functional pillars that form the foundation of the FFEELLAA architecture.



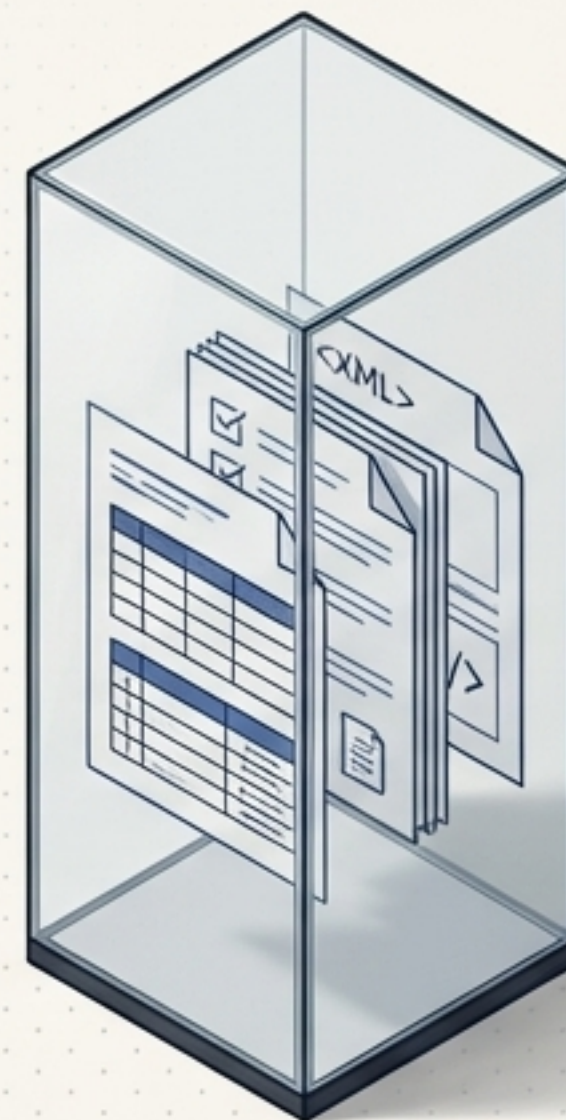
01. Ontology & Rules

Houses the domain knowledge.
Contains Core & Rule ontologies formatted
as Turtle (.ttl) files within /ontologies/.



02. Capabilities Engine

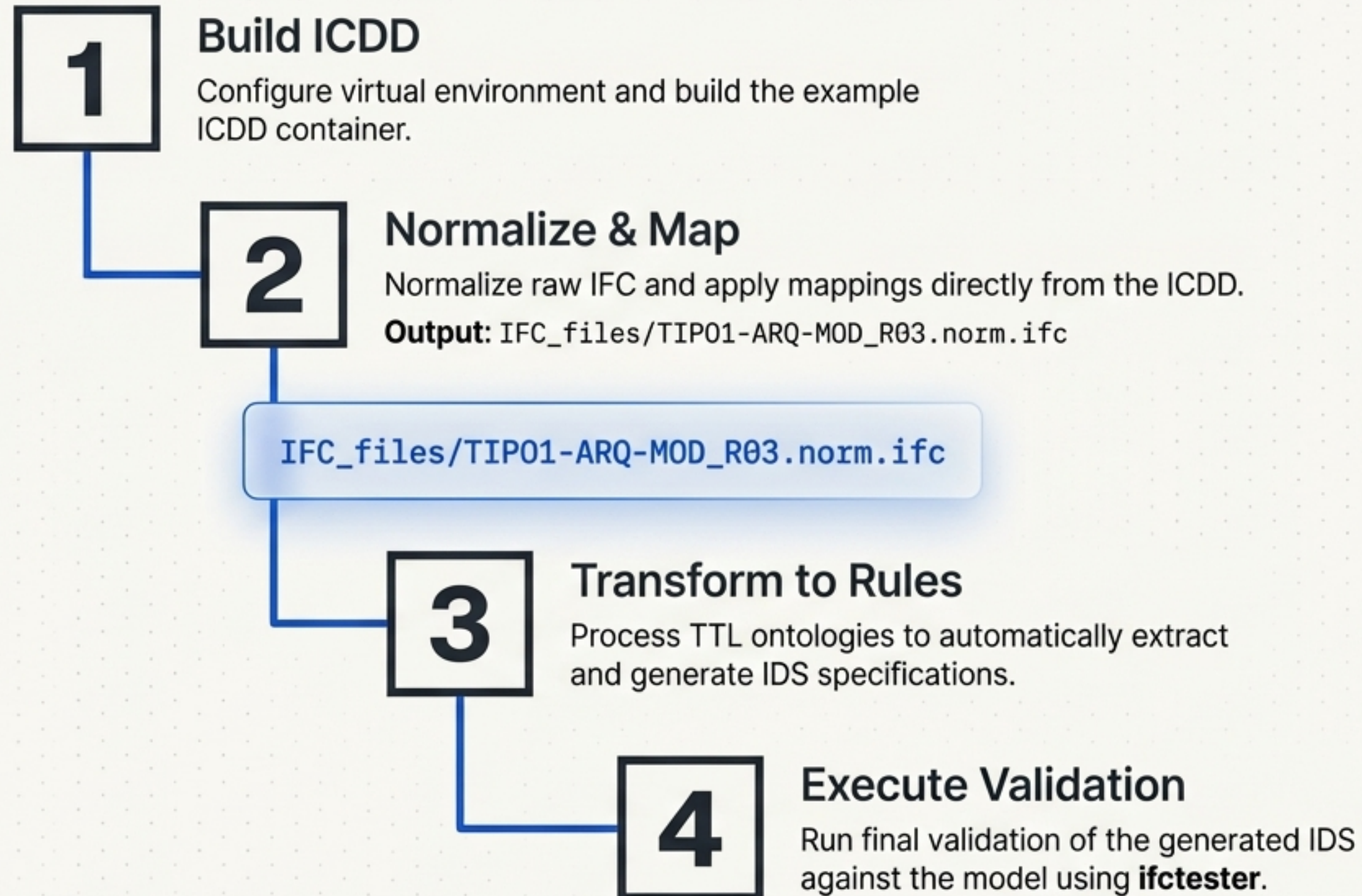
The operational muscle.
Handles ICDD/IFC conversion alongside
the TTL to IDS extraction pipeline.



03. IDS / IDCS Outputs

The formal rule deliverables.
Verifiable artifacts generated from the
capabilities layer, stored in /ids/

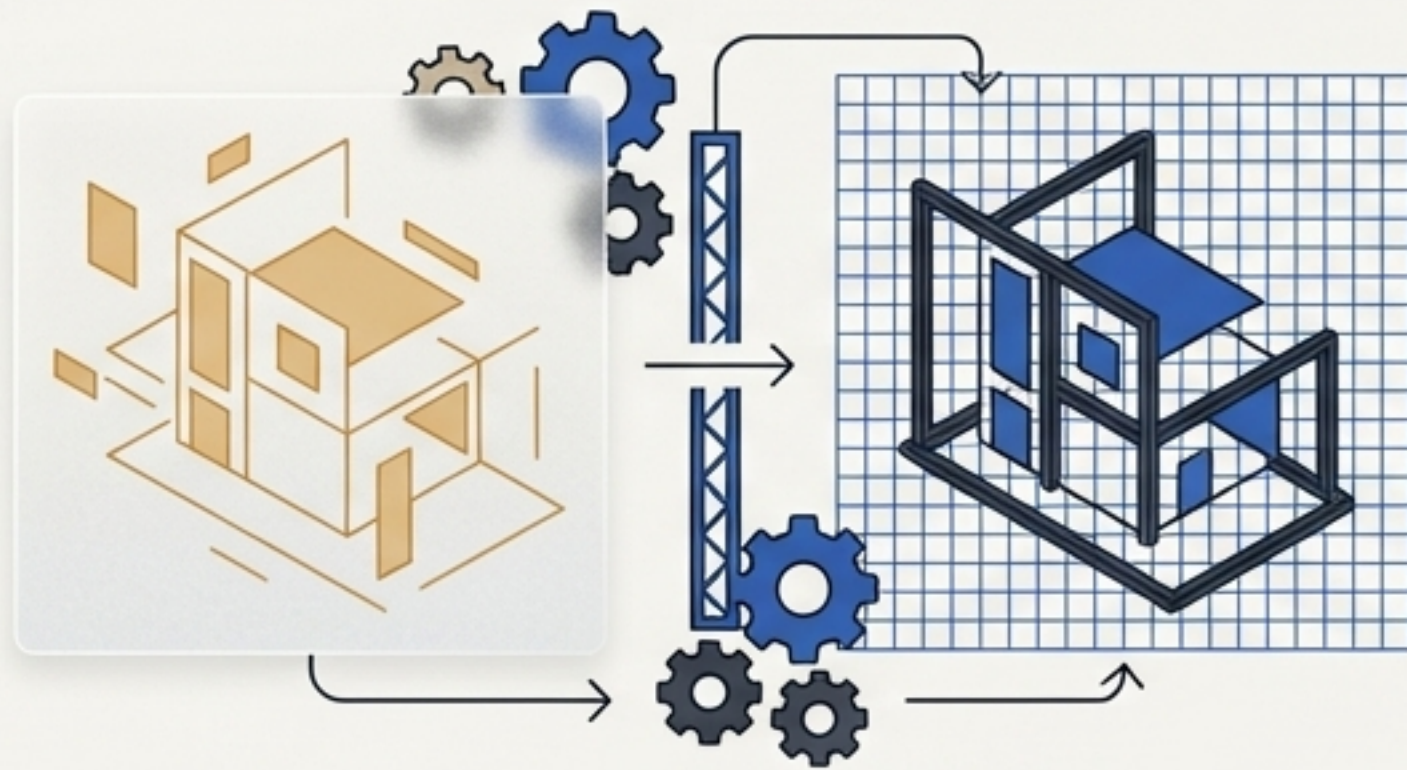
System Activation Sequence



Deep Dive: The Capability Scripts

ICDD Normalization

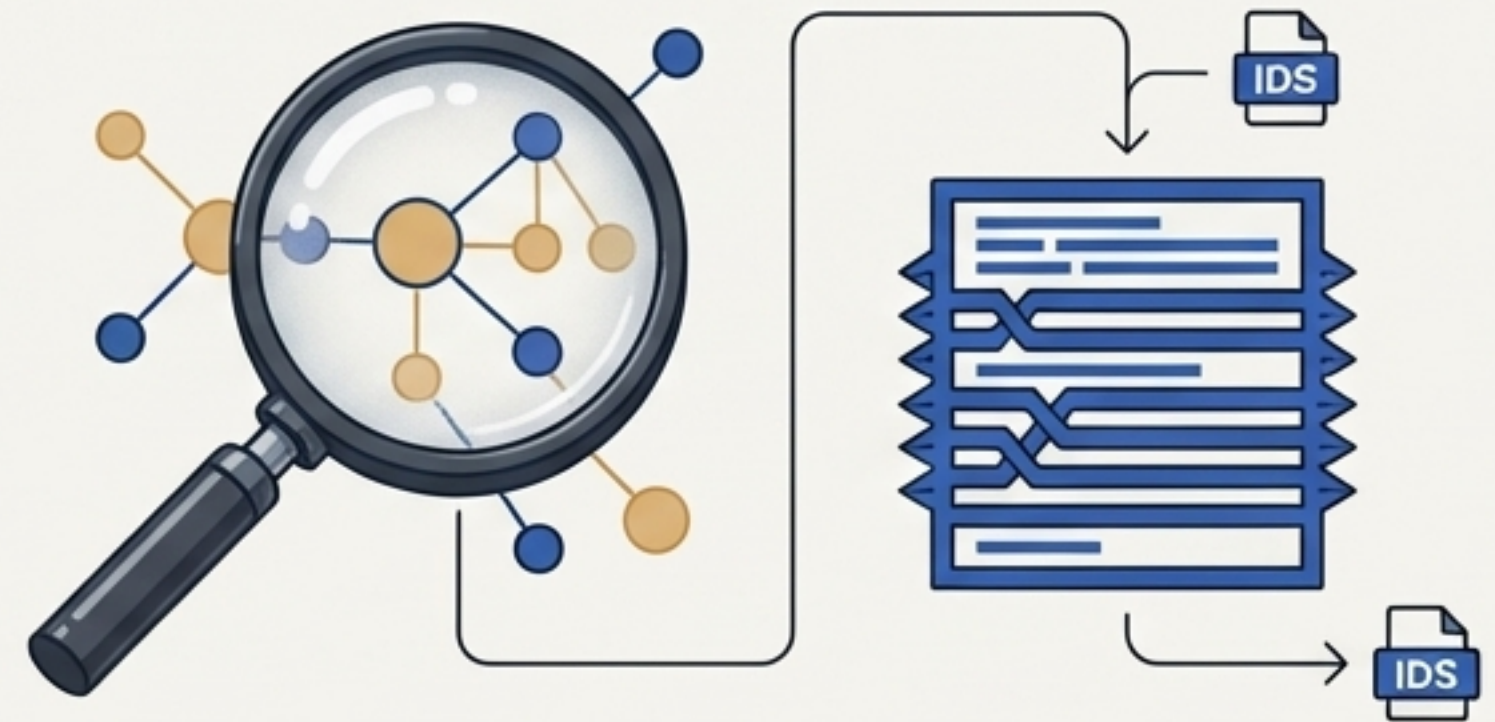
Path: `convert_icdd_ontology.py`



Function: Ingests the raw model, applies ICDD mappings, and normalizes the IFC structural data to prepare it for rule validation.

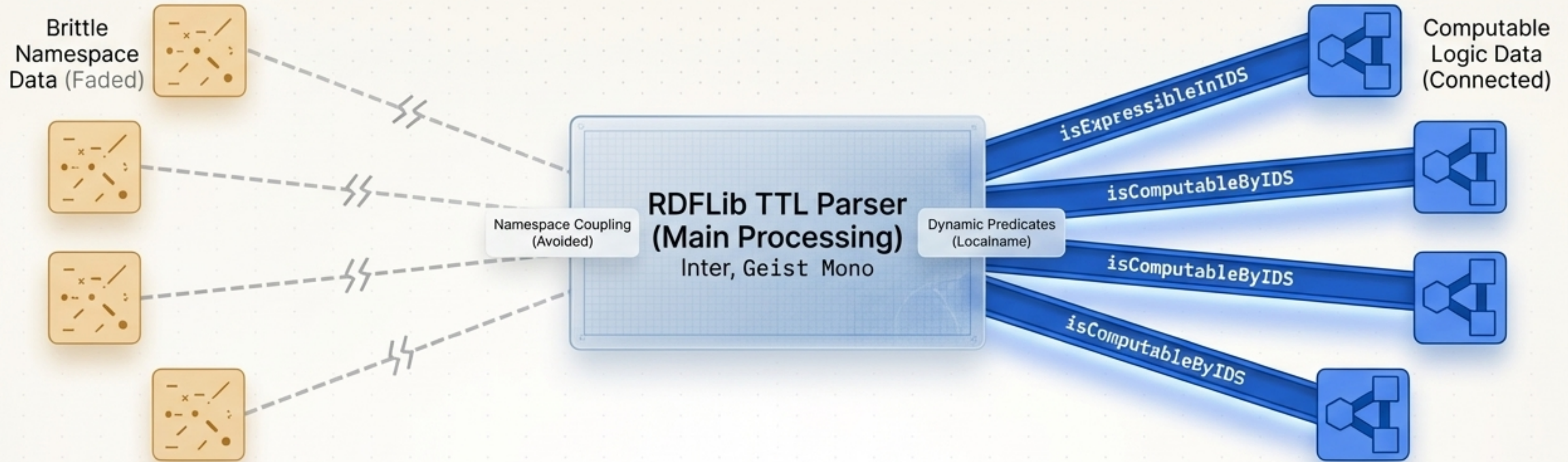
Rule Extraction

Path: `ttrl_to_ids.py`



Function: Parses Turtle files using RDFLib, filters specifically for IDS-expressible rules, and automatically generates the target specifications.

Technical Elegance: Decoupled Parsing



RDFLib & Decoupling

The TTL parser explicitly avoids brittle namespace coupling.

Intelligent Predicate Matching

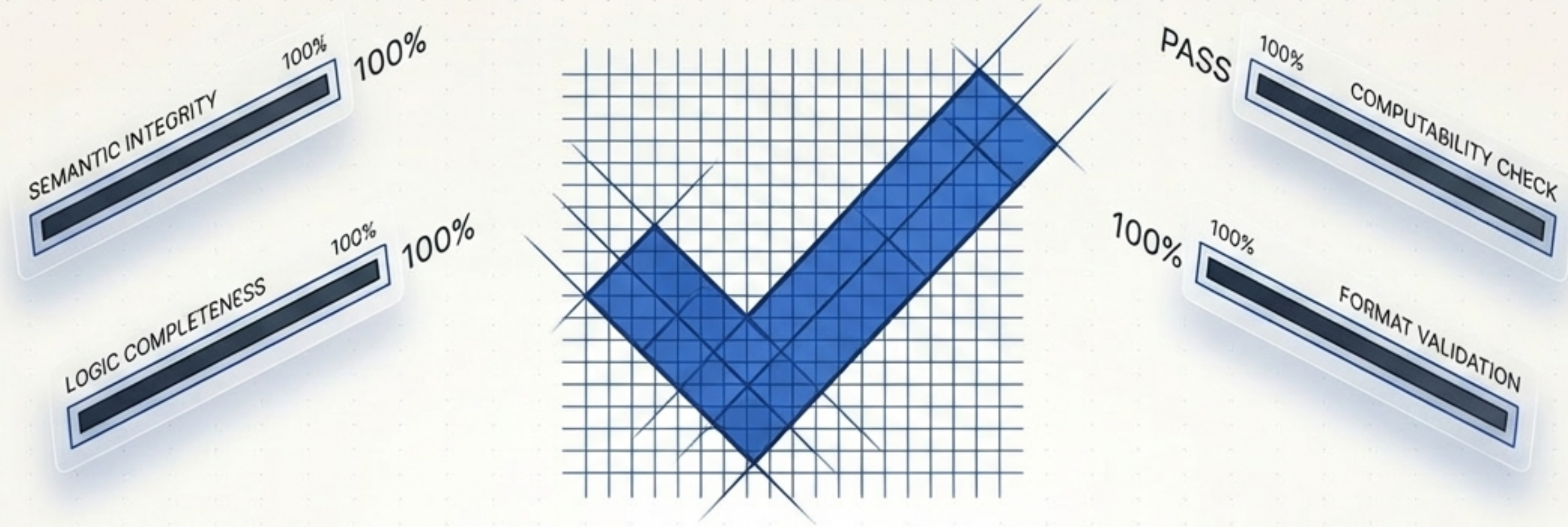
Filters for computable logic by matching predicates dynamically via localname `isExpressibleInIDS` (fallback: `isComputableByIDS`).

Seamless XML Generation

Final artifact generation is natively handled via `ifctester.ids` utilizing `Ids.to_xml(...)`.

Ensuring Extraction Reliability

The critical path of translating semantic ontologies into verifiable logic is strictly validated.



Inter

Coverage ensures that only perfectly formatted, computable rules are passed to the execution layer.

Inter
Test Target: TTL Rule Extractor

Test Suite: `test_ttl_to_ids_read_fela_computable_rules.py`

The Unified Architecture

By decoupling parsing logic, mapping human intent via LLM to structured Turtle ontologies, and utilizing robust **IFC normalization**, FFEELLAA creates an unbreakable chain from initial requirement to final structural validation.

